This note derives the camera pose covariance ${}^{\mathrm{C}}\boldsymbol{\Sigma}_{\mathrm{C}}$ of `pose_estimation.pdf` and the estimate of the pixel noise it is scaled with. For N detected marker corners, the image Jacobian $\boldsymbol{J}$ stacks the 2×6 Jacobians of all corners into a $2N \times 6$ matrix, so each row belongs to one pixel coordinate, u or v, of one corner.

Covariance of the fitted pose. Let $\boldsymbol{\xi}$ be the true camera pose and $\boldsymbol{f}(\boldsymbol{\xi})$ the projection of the marker points into the image at that pose, which maps a camera pose to the stacked $2N$ pixel coordinates of the marker corners. We model the detected corner coordinates $\boldsymbol{m} \in \mathbb{R}^{2N}$ as this projection plus independent pixel noise of equal variance σ^2 on every coordinate,

$$\boldsymbol{m} = \boldsymbol{f}(\boldsymbol{\xi}) + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_{2N}).$$

SQPnP returns the pose $\hat{\boldsymbol{\xi}}$ that minimizes the reprojection cost $E = \frac{1}{2}\|\boldsymbol{r}\|^2$ of the residuals $\boldsymbol{r} = \boldsymbol{m} - \boldsymbol{f}(\hat{\boldsymbol{\xi}})$. Let $\delta\boldsymbol{\xi}$ be the small perturbation that moves the true pose $\boldsymbol{\xi}$ to the estimate $\hat{\boldsymbol{\xi}}$. Linearizing $\boldsymbol{f}$ around the true pose, $\boldsymbol{f}(\hat{\boldsymbol{\xi}}) \approx \boldsymbol{f}(\boldsymbol{\xi}) + \boldsymbol{J}\,\delta\boldsymbol{\xi}$, gives the residuals

$$\boldsymbol{r} \approx \boldsymbol{\varepsilon} - \boldsymbol{J}\,\delta\boldsymbol{\xi}. \tag{1}$$

At the minimum, the gradient of the cost, $\partial E / \partial \delta\boldsymbol{\xi} = -\boldsymbol{J}^\top \boldsymbol{r}$, vanishes. $\delta\boldsymbol{\xi}$ therefore solves the normal equations $\boldsymbol{J}^\top \boldsymbol{J}\,\delta\boldsymbol{\xi} = \boldsymbol{J}^\top \boldsymbol{\varepsilon}$, the linear least-squares fit of a camera motion to the noise,

$$\delta\boldsymbol{\xi} = (\boldsymbol{J}^\top \boldsymbol{J})^{-1} \boldsymbol{J}^\top \boldsymbol{\varepsilon}. \tag{2}$$

Its covariance is

$${}^{\mathrm{C}}\boldsymbol{\Sigma}_{\mathrm{C}} = \mathbb{E}[\delta\boldsymbol{\xi}\,\delta\boldsymbol{\xi}^\top] = (\boldsymbol{J}^\top \boldsymbol{J})^{-1} \boldsymbol{J}^\top\, \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top]\, \boldsymbol{J} (\boldsymbol{J}^\top \boldsymbol{J})^{-1} = \sigma^2 (\boldsymbol{J}^\top \boldsymbol{J})^{-1},$$

OpenCV orders the Jacobian columns as (rotation, translation). We reorder them to (translation, rotation), so that ${}^{\mathrm{C}}\boldsymbol{\Sigma}_{\mathrm{C}}$ has the layout of a ROS pose covariance.

Estimating the pixel noise σ. σ^2 is the variance of a single pixel coordinate, so it is estimated from the $2N$ coordinate residuals $\boldsymbol{r}$. Substituting (2) into (1) gives

$$\boldsymbol{r} = (\boldsymbol{I}_{2N} - \boldsymbol{P})\,\boldsymbol{\varepsilon}, \qquad \boldsymbol{P} = \boldsymbol{J}(\boldsymbol{J}^\top \boldsymbol{J})^{-1} \boldsymbol{J}^\top.$$

The column space of $\boldsymbol{J}$ holds the corner displacements $\boldsymbol{J}\,\delta\boldsymbol{\xi}$ that a small camera motion produces. $\boldsymbol{P}$ is the orthogonal projection onto it: $\boldsymbol{P}\boldsymbol{\varepsilon}$ is the part of the noise that a camera motion explains, and the residuals are the rest, orthogonal to every camera motion.

$\boldsymbol{P}$ is symmetric and idempotent, $\boldsymbol{P}^2 = \boldsymbol{J}(\boldsymbol{J}^\top \boldsymbol{J})^{-1} \boldsymbol{J}^\top \boldsymbol{J} (\boldsymbol{J}^\top \boldsymbol{J})^{-1} \boldsymbol{J}^\top = \boldsymbol{P}$, so $\boldsymbol{I}_{2N} - \boldsymbol{P}$ is too,

$$(\boldsymbol{I}_{2N} - \boldsymbol{P})^\top (\boldsymbol{I}_{2N} - \boldsymbol{P}) = (\boldsymbol{I}_{2N} - \boldsymbol{P})^2 = \boldsymbol{I}_{2N} - 2\boldsymbol{P} + \boldsymbol{P}^2 = \boldsymbol{I}_{2N} - \boldsymbol{P}. \tag{3}$$

For any matrix $\boldsymbol{M}$, the noise coordinates are uncorrelated, $\mathbb{E}[\varepsilon_i \varepsilon_j] = 0$ for $i \neq j$, and $\mathbb{E}[\varepsilon_i^2] = \sigma^2$, so only the diagonal of $\boldsymbol{M}$ remains,

$$\mathbb{E}[\boldsymbol{\varepsilon}^\top \boldsymbol{M} \boldsymbol{\varepsilon}] = \sum_{i,j} M_{ij}\, \mathbb{E}[\varepsilon_i \varepsilon_j] = \sigma^2 \sum_i M_{ii} = \sigma^2 \operatorname{tr}(\boldsymbol{M}). \tag{4}$$

The expected residual sum of squares follows from (3), (4) and the cyclic property of the trace, $\operatorname{tr}(\boldsymbol{AB}) = \operatorname{tr}(\boldsymbol{BA})$,

$$\begin{aligned} \mathbb{E}[\boldsymbol{r}^\top \boldsymbol{r}] &= \mathbb{E}[\boldsymbol{\varepsilon}^\top (\boldsymbol{I}_{2N} - \boldsymbol{P})^\top (\boldsymbol{I}_{2N} - \boldsymbol{P}) \boldsymbol{\varepsilon}] \overset{(3)}{=} \mathbb{E}[\boldsymbol{\varepsilon}^\top (\boldsymbol{I}_{2N} - \boldsymbol{P}) \boldsymbol{\varepsilon}] \overset{(4)}{=} \sigma^2 \operatorname{tr}(\boldsymbol{I}_{2N} - \boldsymbol{P}) \\ &= \sigma^2 \left(2N - \operatorname{tr}\left((\boldsymbol{J}^\top \boldsymbol{J})^{-1} \boldsymbol{J}^\top \boldsymbol{J}\right)\right) = (2N - 6)\, \sigma^2. \end{aligned} \tag{5}$$

Dividing the residual sum of squares by $2N - 6$ therefore gives an unbiased estimate of σ^2, $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$ [2], with

$$\hat{\sigma}^2 = \frac{\boldsymbol{r}^\top \boldsymbol{r}}{2N - 6} = \frac{1}{2N - 6} \sum_{i=1}^{N} \left(\Delta u_i^2 + \Delta v_i^2\right), \tag{6}$$

which needs $N \geq 4$ corners. A single marker always provides 4.

The 6 in (5) is the trace of $\boldsymbol{P}$, the number of pose parameters. Fitting them makes the residuals satisfy the 6 normal equations $\boldsymbol{J}^\top \boldsymbol{r} = \boldsymbol{0}$, so only $2N - 6$ of them vary freely. Fitting a mean to n samples is the case $\boldsymbol{J} = [1, \ldots, 1]^\top$: $\boldsymbol{P}$ replaces every sample with the mean, $\operatorname{tr}(\boldsymbol{P}) = 1$, the normal equation states that the residuals sum to zero, and (6) becomes the sample variance with its $n - 1$.

Lower bound. With one marker, $2N - 6 = 2$: the normal equations leave the 8 residuals only 2 independent components, so $\hat{\sigma}^2$ is (in effect) an average of just two squared noise samples. Two samples are too few to average out (in about 10 % of the frames, $\hat{\sigma}^2$ is below a tenth of the true variance[1]). A localization filter would trust such a near-zero covariance blindly. The pixel noise should therefore be floored at the parameter $\sigma_{\min}$,

$$\sigma^2 = \max\left(\hat{\sigma}^2, \sigma_{\min}^2\right), \qquad {}^{\mathrm{C}}\boldsymbol{\Sigma}_{\mathrm{C}} \approx \sigma^2 (\boldsymbol{J}^\top \boldsymbol{J})^{-1}.$$

The noise of a static camera's corner detections, or the RMS error of the camera calibration, gives a value for $\sigma_{\min}$. The residuals capture the errors that make the markers inconsistent with each other, such as corner bias, a mis-surveyed marker or calibration errors, and they change with the blur, lighting and distance in each frame. Errors that shift all markers consistently, such as an error in ${}^{\mathrm{B}}\boldsymbol{H}_{\mathrm{C}}$, leave no residual and are not modeled.

[1] To first order, (1) is a linear regression model with design matrix $\boldsymbol{J}$, so $\boldsymbol{r}^\top \boldsymbol{r}/\sigma^2 = (2N - 6)\, \hat{\sigma}^2/\sigma^2$ follows a χ^2 distribution with $2N - 6$ degrees of freedom [1]. For one marker, that is χ_2^2, the exponential distribution with mean 2 [3], so $P(\hat{\sigma}^2 < k\sigma^2) = P(\chi_2^2 < 2k) = 1 - e^{-k}$.

References

[1] The Book of Statistical Proofs, "Distribution of residual sum of squares in multiple linear regression", https://statproofbook.github.io/P/mlr-rssdist.

[2] The Book of Statistical Proofs, "Maximum likelihood estimator of variance in multiple linear regression is biased", https://statproofbook.github.io/P/resvar-biasp.

[3] Wikipedia, "Chi-squared distribution", section "Related distributions", https://en.wikipedia.org/wiki/Chi-squared_distribution.